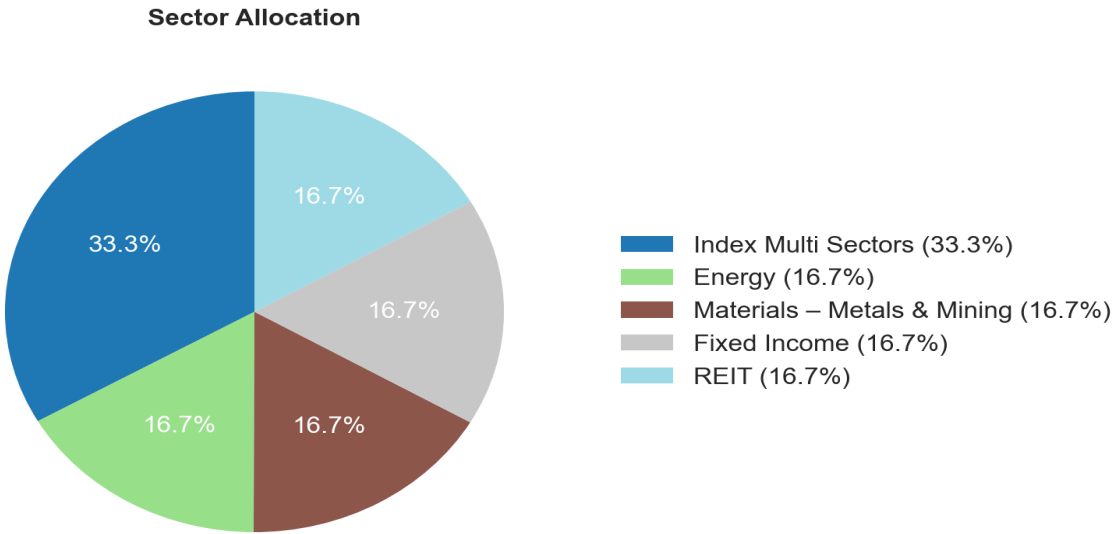


Portfolio rotation report

Date of extraction	2025-12-07 23:31
Data window	2019-12-30 → 2024-10-21
Frequency	Weekly — 252 observations. Sufficient depth: short- and long-term indicators are statistically reliable. Warning: the observation window exceeds the typical interpretability range for this frequency (+384.6% above the recommended upper bound). Results may blend multiple market regimes, reducing economic signal clarity.
Tickers	GLD, QQQ, SPY, TLT, VNQ, XLE
Indicators	Short term: 8 weeks volatility, 12 weeks return, 16 weeks momentum Long term: 20 weeks volatility, 24 weeks return, 48 weeks momentum

1. Portfolio Composition

- XLE (Energy Select Sector SPDR Fund — Energy)
 - GLD (SPDR Gold Shares — Materials – Metals & Mining)
 - TLT (iShares 20+ Year Treasury Bond ETF — Fixed Income)
- SPY (SPDR S&P; 500 ETF — Index Multi Sectors)
 - VNQ (Vanguard Real Estate ETF — REIT)
 - QQQ (Invesco QQQ Trust — Index Multi Sectors)



Macro Profile Insight

Macro coverage

- Partial coverage: several macro themes remain unrepresented.
- Broad balance across market capitalizations: diversification achieved across the size spectrum.

Structural properties

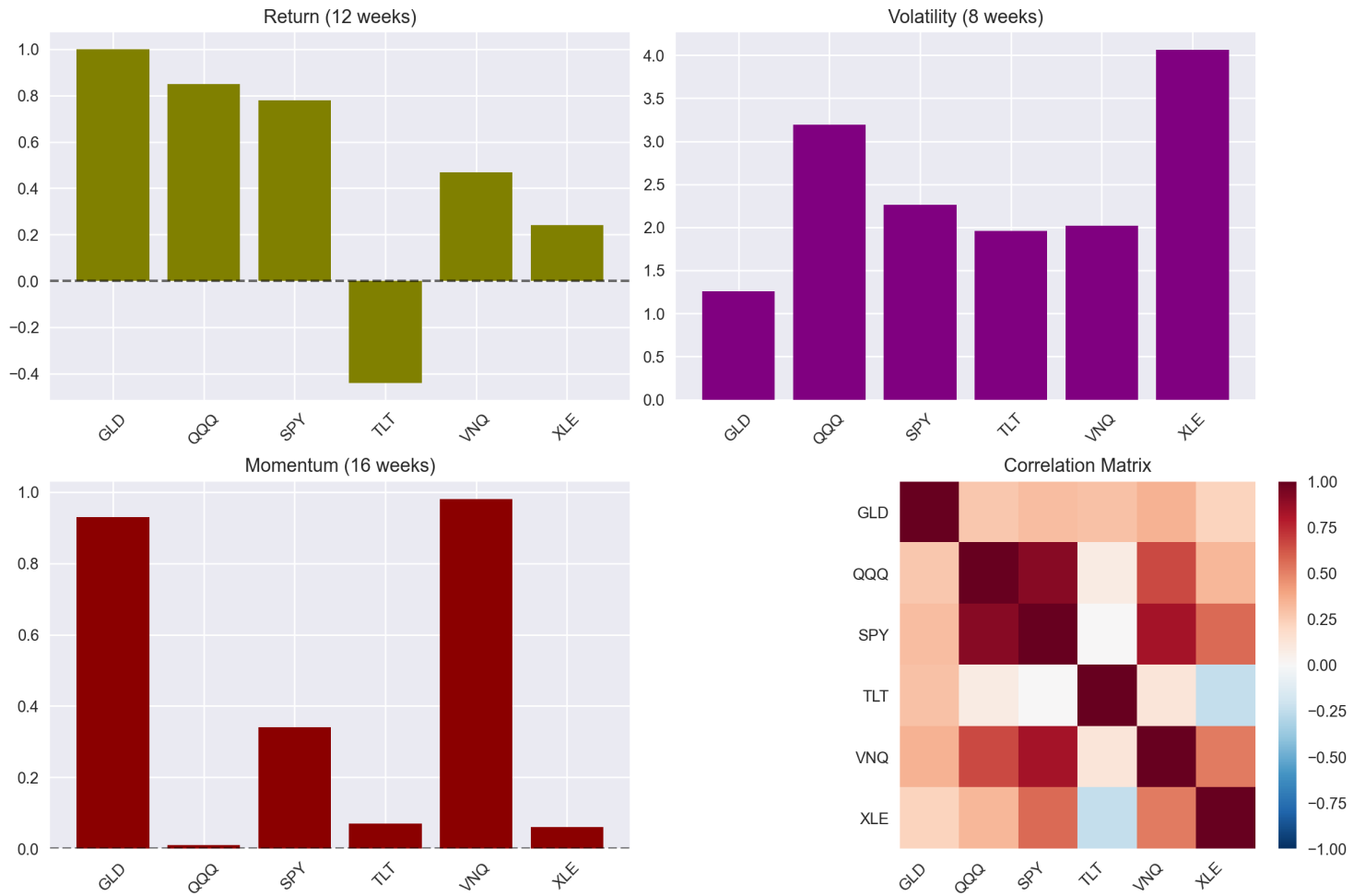
- Concentration risk: portfolio heavily exposed to etf instruments.
- Liquidity stance: neutral liquidity conditions overall; robust median dollar volume.
- Defensive beta tilt: lower systematic risk profile.
- Balanced volatility structure: risk evenly distributed across holdings.

Additional comments

- Significant sector gaps: broad portions of the economy remain uncovered.

2. Sector Performance Visualization

Sector Performance Dashboard



Top Positive Correlations

66.7% of portfolio

Pair	Correlation
QQQ / SPY	0.91
SPY / VNQ	0.83
QQQ / VNQ	0.66
SPY / XLE	0.57
VNQ / XLE	0.52

Top Neutral Correlations

33.3% of portfolio

Pair	Correlation
SPY / TLT	0.0
QQQ / TLT	0.08
TLT / VNQ	0.12
GLD / XLE	0.22
TLT / XLE	-0.24

Overall, average cross-sector correlation was 0.35, ranging from -0.24 to 0.91. This suggests elevated co-movement — diversification benefits are partially reduced. Correlation dispersion is wide, showing sharp contrasts between sectors. Strongest co-movement between QQQ and SPY (0.91); weakest between TLT and XLE (-0.24). Mixed regime: both market-wide and sector-specific forces are at play.

3. Key Performance Indicators

3.a Current Regime Snapshot

Ticker	R (%)	V (%)	RAR	M (%)	Signal	Regime	Liquidity	Commentary
GLD	1.00	1.26	0.797	0.93	mildly coherent	elevated	thin	positive performance with solid momentum, signal remains coherent, but with limited conviction, rising volatility — tension building, under limited liquidity.
QQQ	0.85	3.19	0.266	0.01	neutral	subdued	liquid	mixed short-term dynamics, signal unclear or in transition, low volatility — calm market tone, with ample liquidity.
SPY	0.78	2.26	0.346	0.34	mildly coherent	subdued	liquid	mixed short-term dynamics, signal remains coherent, but with limited conviction, low volatility — calm market tone, with ample liquidity.
TLT	-0.44	1.96	-0.223	0.07	neutral	subdued	liquid	recovering momentum despite recent losses, signal unclear or in transition, low volatility — calm market tone, with ample liquidity.
VNQ	0.47	2.02	0.232	0.98	mildly coherent	subdued	illiquid	mixed short-term dynamics, signal remains coherent, but with limited conviction, low volatility — calm market tone, under limited liquidity.
XLE	0.24	4.06	0.06	0.06	neutral	subdued	liquid	mixed short-term dynamics, signal unclear or in transition, low volatility — calm market tone, with ample liquidity.

Top by Average Return

Ticker	AvgReturn(%)
GLD	1.0
QQQ	0.85
SPY	0.78
VNQ	0.47
XLE	0.24

Top by Volatility

Ticker	Volatility(%)
XLE	4.06
QQQ	3.19
SPY	2.26
VNQ	2.02
TLT	1.96

Top by Momentum

Ticker	Momentum(%)
VNQ	0.98
GLD	0.93
SPY	0.34
TLT	0.07
XLE	0.06

Last by Average Return

Ticker	AvgReturn(%)
TLT	-0.44
XLE	0.24
VNQ	0.47
SPY	0.78
QQQ	0.85

Last by Volatility

Ticker	Volatility(%)
GLD	1.26
TLT	1.96
VNQ	2.02
SPY	2.26
QQQ	3.19

Last by Momentum

Ticker	Momentum(%)
QQQ	0.01
XLE	0.06
TLT	0.07
SPY	0.34
GLD	0.93

3.b Rotation & Stress Metrics

Ticker	ΔReturn (pp)	Momentum Ratio	Signal	Volatility Ratio	Beta	ADV10 USD	Interpretation
GLD	0.39	1.21	0.421	0.95	0.04	9.35 M	stable performance, momentum stable, volatility normalizing, aligned with market dynamics, under robust trading volume.
QQQ	0.39	0.02	0.01	1.10	1.18	53.05 M	stable performance, momentum fading, volatility normalizing, aligned with market dynamics, under robust trading volume.
SPY	0.30	0.55	0.22	1.20	1.00	72.91 M	stable performance, momentum fading, volatility normalizing, aligned with market dynamics, under robust trading volume.
TLT	-0.59	0.51	0.046	0.88	0.07	34.04 M	stable performance, momentum fading, volatility normalizing, aligned with market dynamics, under robust trading volume.
VNQ	-0.20	1.87	0.341	1.14	0.59	3.60 M	stable performance, momentum surging, volatility normalizing, aligned with market dynamics, under robust trading volume.
XLE	0.37	0.29	0.047	1.44	0.80	28.61 M	stable performance, momentum fading, volatility expanding, aligned with market dynamics, under robust trading volume.

■ Moderate-impact observations

For **GLD**: *Stabilization after shock* — volatility normalizes; confirmation still needed. (avg ann. return 1.0%, momentum 0.9%, vol 1.3%)

■ Light-impact observations

For **QQQ, SPY, TLT, VNQ, XLE**: *Volatility compression* — subdued volatility; a breakout setup may be forming. (avg ann. return 0.4%, momentum 0.3%, vol 2.7%)